FIITJEE

ALL INDIA TEST SERIES

FULL TEST – XI

JEE (Main)-2022

TEST DATE: 17-04-2022

Time Allotted: 3 Hours **Maximum Marks: 300**

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts.**
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B :** Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) *contains 60 multiple choice questions which have* ***only one correct answer****. Each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Section-B (21 – 30, 51 – 60, 81 – 90) *contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries* ***+4 marks*** *for correct answer and* ***–1 mark*** *for wrong answer.*

Physics

PART – A

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A glass wind screen whose inclination with the vertical can be changed is mounted on a car. The car moves horizontally with a speed of 4 m/s. At what angle α with the vertical should the wind screen be placed so that rain drops falling vertically downwards with velocity 12 m/s strike the wind screen perpendicularly?

   (A) $\tan^{-1}\left(\frac{1}{3}\right)$
   (B) $\tan^{-1}(3)$
   (C) $\cos^{-1}(3)$
   (D) $\sin^{-1}\left(\frac{1}{3}\right)$

2. Find the amount of work done to increase the temperature of one mole of an ideal gas by 30°C if it is expanding under the condition $V \propto T^{2/3}$

   (A) 166.2 J
   (B) 136.2 J
   (C) 126.2 J
   (D) None of these

3. A particle is hanging from a string of length ℓ = 10 m. It is imparted a velocity of 20 m/s at the bottom. The velocity of the particle at its highest point is (g = 10 m/s^2)

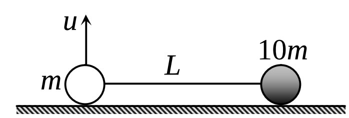
   (A) $\frac{20}{3}$ m/s
   (B) $\frac{20}{3}\sqrt{\frac{2}{3}}$ m/s
   (C) $\sqrt{\frac{20}{3}}$ m/s
   (D) $20\sqrt{\frac{2}{3}}$ m/s

4. Suppose the gravitational force varies inversely as the $\frac{1}{n^{th}}$ power of distance. Then the time period of a planet in circular orbit of radius *R* around the sun will be proportional to

   (A) $R^{\left(\frac{n+1}{2n}\right)}$
   (B) $R^{\left(\frac{n-1}{2}\right)}$
   (C) R^{n}
   (D) $R^{\left(\frac{n-2}{2}\right)}$

5. A ball of mass *m* and density ρ is immersed in a liquid of density 5ρ at a depth *h* and released. To what height will the ball jump up above the surface of liquid? (neglect the resistance of water and air).
(A) h
(B) h/2
(C) 4h
(D) 3h

6. As shown in the figure, a mass *m* and another mass 10*m* are connected with a string. Friction is sufficient to prevent the slipping of 10*m*. Mass *m* is given a velocity *u* in vertical direction. For complete circular motion of mass *m* keeping heavier particle stationary, the value of *u* is



(A) $u > \sqrt{3gL}$
(B) $\sqrt{2gL} < u < \sqrt{5gL}$
(C) $\sqrt{3gL} < u < \sqrt{13gL}$
(D) $\sqrt{11gL} < u < \sqrt{15gL}$

7. If the length of a simple pendulum is equal to 2R, where R is the radius of earth, its time period of S.H.M. will be
(A) $2\pi\sqrt{\frac{2R}{3g}}$
(B) $2\pi\sqrt{\frac{R}{2g}}$
(C) $2\pi\sqrt{\frac{2R}{g}}$
(D) $\pi\sqrt{\frac{R}{2g}}$

8. An organ pipe of length 33 cm closed at one end vibrates in its 5th overtone. If amplitude of a particle at antinodes is 6 mm, then amplitude of a particle which is at a distance 18 cm from closed end is
(A) 3 cm
(B) $3\sqrt{2}$ mm
(C) 2 mm
(D) zero

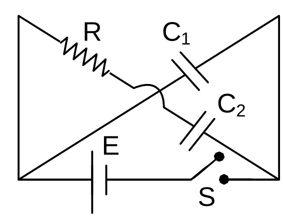
9. A uniform ring of radius R is given a back spin of angular velocity $\frac{v_0}{2R}$ and thrown on a horizontal rough surface with velocity of centre to be v_0. The velocity of the centre of the ring when it starts pure rolling will be
(A) $\frac{v_0}{2}$
(B) $\frac{v_0}{4}$
(C) $\frac{3v_0}{4}$
(D) 0

10. The depletion layer is silicon diode is 1 μm wide and the knee potential is 0.6 V, then the electric field in depletion layer will be
    (A) zero
    (B) 0.6 V/m
    (C) 6×10^4 V/m
    (D) 6×10^5 V/m

11. The wavelength corresponding to maximum spectral radiancy of a black body A is λ_A = 5000Å. Consider another black body B, whose surface area is twice that of A and total radiant energy by B is 16 times that emitted by A in same time. The wavelength corresponding to maximum spectrum radiancy for B will be
    (A) $5000(2)^{3/4}$ Å
    (B) 2500 Å
    (C) 10000 Å
    (D) $5000(2)^{-3/4}$ Å

12. The wavelength of characteristic K_α line emitted by a hydrogen like element is 0.32 Å. The wavelength of the K_β line emitted by the same element will be
    (A) 0.25 Å
    (B) 0.27Å
    (C) 0.30Å
    (D) 0.35Å

13. In the circuit shown, the capacitors C_1 and C_2 have capacitance C each. The switch S is closed at time t = 0. Taking Q_0 = CE and τ = RC, the charge on C_2 after time t will be



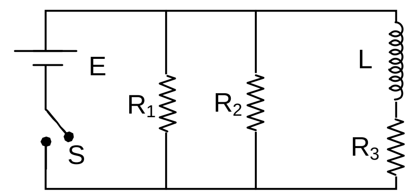
    (A) $Q_0\left(1-e^{-t/2\tau}\right)$
    (B) $Q_0\left(1-e^{-t/\tau}\right)$
    (C) $\frac{Q_0}{2}\left(1-e^{-t/2\tau}\right)$
    (D) $Q_0\left(1-e^{-2t/\tau}\right)$

14. The output of an AC generator is given by:
    $E = E_m \sin\left(\omega t - \frac{\pi}{4}\right)$ and current is given by $i = i_m \sin\left(\omega t - \frac{3\pi}{4}\right)$. The circuit contains a single element other than the generator. It is:
    (A) a capacitor
    (B) a resistor
    (C) an inductor
    (D) not possible to decide due to lack of information

15. A particle undergoes SHM with a time period of 2 second. In how much time will it travel from its mean position to a displacement equal to half of its amplitude
    (A) $\frac{1}{2}$ sec
    (B) $\frac{1}{3}$ sec
    (C) $\frac{1}{4}$ sec
    (D) $\frac{1}{6}$ sec

16. Two beams of light having intensities I and 4I interfere to produce a fringe pattern on a screen. The phase difference between the beams is $\pi/2$ at point A and π at point B on the screen. Then the difference between resultant intensities at A and B is:
(A) 2I
(B) 4I
(C) 5I
(D) 7I

17. An inductor of inductance L = 400 mH and three resistor of resistance $R_1 = 4\Omega$, $R_2 = 2\Omega$ and $R_3 = 10\Omega$ are connected to a battery of emf E = 10V as shown in the figure. The internal resistance of the battery is negligible. The switch S is closed at tim t = 0. What is the potential drop across L as a function of time?

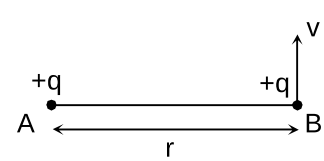


(A) $10e^{-25t}$ volt
(B) $10e^{-5t}$
(C) $5e^{-5t}$ volt
(D) $30e^{-16t}$ volt

18. 3.2 kg of ice at –10°C just melts when with a mass m of steam, at 100°C is passed through it. Then (specific heat of ice = 0.5 cal/gm°C, latent heat of ice = 80 cal/gm, latent heat of vaporization of water = 540 cal/gm, specific heat of water = 1 cal/gm°C
(A) m = 400 g
(B) m = 800 g
(C) m = 425 g
(D) m = 900 g

19. To get maximum current through a resistance of 2.5 Ω, one can use 'm' rows of cells, each row having 'n' cells. The internal resistance of each cell is 0.5Ω. What are the values of n and m, if the total number of cells is 45.
(A) 3, 15
(B) 5, 9
(C) 9, 5
(D) 15, 3

20. In the figure shown, A is a fixed charge. B (of mass m) is given a velocity V perpendicular to line AB. At this moment the radius of curvature of the resultant path of B is



(A) 0
(B) ∞(infinity)
(C) $\dfrac{4\pi\varepsilon_0 r^2 mv^2}{q^2}$
(D) r

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

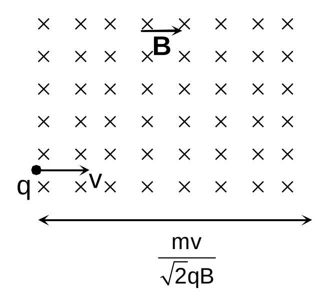
21. A uniform cube of side *a* and mass *m* rests on a rough horizontal table. A horizontal force *F* is applied normal to one of the faces at a point that is directly above the centre of face, at a height of $\frac{4}{5}a$ above the base. The minimum value of *F* for which the cube will begin to tilt about an edge is $\frac{kmg}{4}$. Find the value of k. (assume that the cube does not slide)

22. A ring of mass *m* is placed on a rough horizontal surface with its plane vertical. A horizontal impulse *J* is applied on the ring of mass *m* along a line passing through its centre. The linear velocity of the centre of the ring once it starts pure rolling is $\frac{kJ}{m}$. Find the value of k.

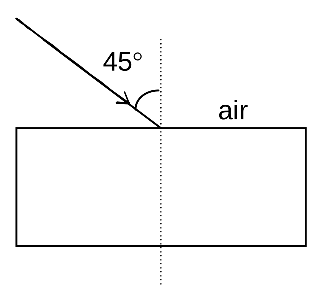
23. A particle of mass 1kg is performing SHM with maximum kinetic energy 2J. The average speed of particle during the interval of time in which it moves from one extreme position to the other extreme position is $\frac{n}{\pi}$ m/s. Fin the value of n.

24. Two bodies of mass 1 kg and 2 kg move towards each other in mutually perpendicular direction with the velocities 3 m/s and 2 m/s respectively. If the bodies stick together after collision what will be the value of heat liberated in joule.

25. A positive charge q is projected in magnetic field of width $\frac{mv}{\sqrt{2}qB}$ with velocity v as shown in the figure. Then time taken by charged particle to emerge from the magnetic field is $\frac{k\pi m}{qB}$. Find the value of k.

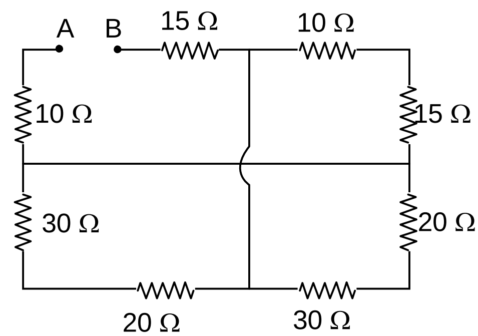


26. A ray of light enters into a transparent liquid from air as shown in the figure. The refractive index of the liquid varies with depth x from the topmost surface as $\mu = \sqrt{2} - \frac{1}{\sqrt{2}}x$ where x in meters. The depth of the liquid medium is sufficiently large. Find the maximum depth (in cm) reached by the ray inside the liquid.



27. A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass number 80 and 120. If binding energy per nucleon for parent atom is 6.5 MeV and for daughter nuclei is 7 MeV and 8 MeV respectively, then the energy released in the decay will be k $\times 10^2$ MeV. Find k.

28. The equivalent resistance of the circuit across points A and B is nΩ. Find the value of n.



29. A paramagnetic substance of susceptibility 3×10^{-4} placed in a magnetic field of 4×10^{-4} Am^{-1}. Then, the intensity of magnetization in the unit of Am^{-1} is $k \times 10^{-8}$. Find the value of k.

30. The maximum kinetic energy of photo electrons emitted from a metal surface, when photon of energy 6eV fall on it is 4 eV. Find the stopping potential in volts.

Chemistry

PART – B

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

31. The de-Broglie's wavelength of electron in a certain orbit of Li^{2+} is $2\pi a_o \left[a_o = 0.53\,\overset{o}{A}\right]$. What is the quantum number of that orbit?
(A) 2
(B) 3
(C) 4
(D) 1

32. 24 miligram of $MgSO_4$ is present in 500 ml of a water sample. What is the hardness of water in ppm?
(A) 60 ppm
(B) 20 ppm
(C) 40 ppm
(D) 50 ppm

33. Which of the following statement is incorrect?
(A) The hybridization of Be in solid $BeCl_2$ is sp^3.
(B) In B_2H_6 all of the B – H bonds are not of equal length.
(C) The angle between H – O and O – O bond in hydrogen peroxide is same in gas and in solid.
(D) O_3 is a polar molecule.

34. Which of the following statement is incorrect?
(A) ${}^{2}_{1}H$ and ${}^{3}_{2}He$ are isotones
(B) ${}^{37}_{17}Cl$ and ${}^{39}_{18}Ar$ are isodiaphers
(C) ${}^{7}_{3}Li$ and ${}^{7}_{4}Be$ are isobars
(D) ${}^{3}_{1}He$ and ${}^{4}_{2}He$ are mirror nuclei

35. In which of the following cases, the oxidation sate number changes by 8?
(A) $MnO_4^- \longrightarrow MnO_2$
(B) $C_2O_4^{2-} \longrightarrow CO_2$
(C) $AsO_3^{3-} \longrightarrow AsO_4^{3-}$
(D) $NO_3^- \longrightarrow NH_4^+$

36. 20 ml of $\frac{M}{10}$ H_3PO_4 solution is treated with 40 ml of $\frac{M}{10}$ NaOH solution. What is the pH of the resulting solution? $\left[pK_1 = 3, pK_2 = 7, pK_3 = 13 \text{ for } H_3PO_4\right]$
(A) 10
(B) 9
(C) 5
(D) 8

37. Calculate E^o for $MO_2^+ \longrightarrow M^{+4}$ in 1 M acidic medium in the following diagram

$$\begin{tikzcd} MO^{2+} \arrow[r, "0.97\ V"] \arrow[rrr, bend right, "1.02\ V"'] & MO_2^+ \arrow[r, "?"] & M^{+4} \arrow[r, "1.01\ V"] & M^{+3} \end{tikzcd}$$

(A) 1.20 V
(B) 1.08 V
(C) 1.32 V
(D) 1.5 V

38. In the Lassign's test of an organic compound, if $FeCl_3$ is added in the final extract, a red colouration is observed. Which elements are present in the compound apart from carbon and hydrogen?
(A) N and O
(B) N and P
(C) S and O
(D) S and N

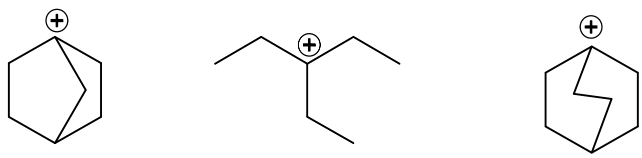
39. 3 mole of nitrogen initially at 300 K and 10 atm expands irreversibly and adiabatically against a constant pressure of 1 atm. Calculate the final temperature of the gas $\left(C_P = 29.133\ K^{-1}\right)$.
(A) 230 K
(B) 270 K
(C) 223 K
(D) 242 K

40. If $K_4\left[Fe(CN)_6\right]$ undergoes 50% dissociation in water, calculate the van't Hoff factor i?
(A) 4
(B) 5
(C) 3
(D) 2

41. Phosphorescence shown by phosphorus is due to
(A) Oxidation of P in P_2O_3
(B) Oxidation of P in P_2O_5
(C) Oxidation of Luciferin by Lucifeze enzyme
(D) Reduction of P into phosphide ion

42. Choose the correct order of Δ_o for the following complexes
I. $\left[Co(H_2O)_6\right]^{2+}$
II. $\left[Co(H_2O)_6\right]^{3+}$
III. $\left[Fe(H_2O)_6\right]^{3+}$
IV. $\left[Fe(CN)_6\right]^{3-}$
(A) I < II < III < IV
(B) I < III < II < IV
(C) I < II = III < IV
(D) I < II < IV < III

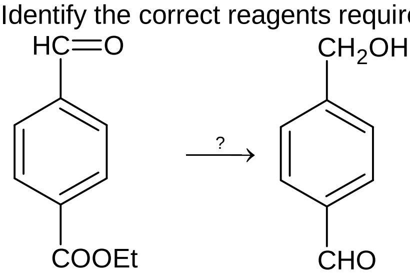
43. Which of the following acetate salts gives the metal on heating?
(A) $Na_2C_2O_4$
(B) $Ag_2C_2O_4$
(C) FeC_2O_4
(D) SnC_2O_4

44. 

I II III

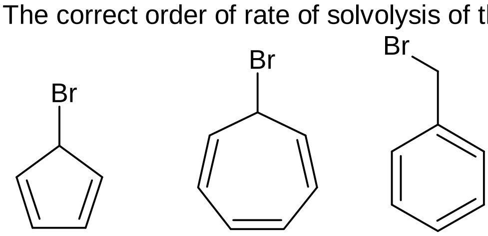
The correct order of stability of the given carbonium ions is
(A) II > I > III
(B) III > II > I
(C) I > III > II
(D) II > III > I

45. Identify the correct reagents required for the following transformation



(A) (i) $NaBH_4$ (ii) H_3O^+
(B) (i) $LiAlH_4$ (ii) H_3O^+
(C) (i) $HOCH_2CH_2-OH$ (ii) $LiAlH_4$ (iii) H_3O^+
(D) (i) $HS-CH_2-CH_2-SH, H^+$ (ii) $LiAlH_4$ (iii) H_3O^+

46. The correct order of rate of solvolysis of the following molecule is



I II III

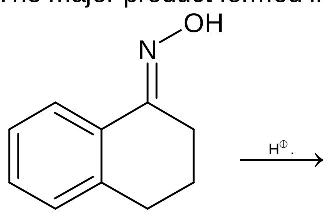
(A) III > II > I
(B) II > I > III
(C) III > I > II
(D) II > III > I

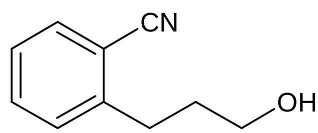
47. The metal ion M^{2+} in the following reaction is

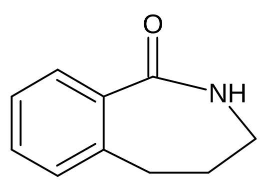
$$M^{2+} + S^{2-} \longrightarrow \text{Black precipitate} \xrightarrow[\text{HNO}_3]{\text{Hot conc.}} \text{White precipitate}$$

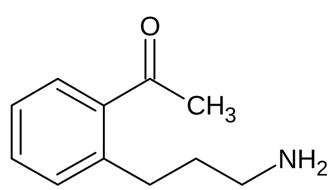
(A) Mn^{2+}
(B) Fe^{2+}
(C) Cd^{2+}
(D) Cu^{2+}

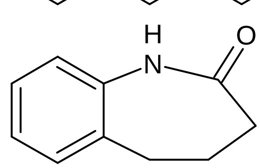
48. The major product formed in the following reaction is



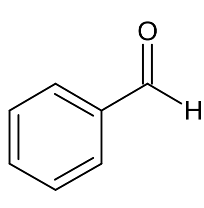
(A)


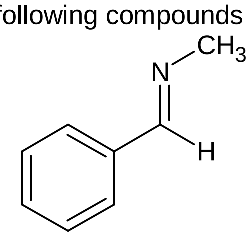
(B)


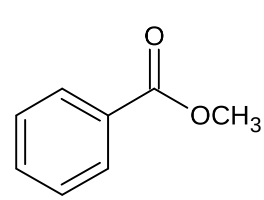
(C)


(D)


49. What is the reactivity of the following compounds towards nucleophile

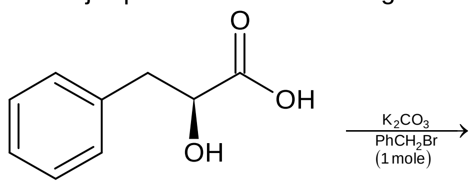

P


Q

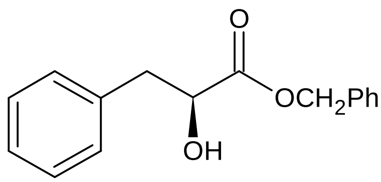

R

(A) P > Q > R
(B) Q > P > R
(C) Q > R > P
(D) R > P > Q

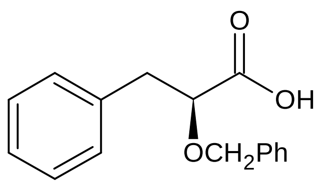
50. The major product in the following reaction



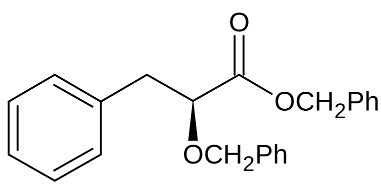
(A)



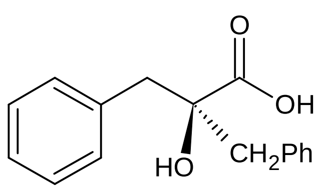
(B)



(C)



(D)



SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

51. For the following amino acids pK_a of the groups are given

$$\underset{pK_a = 4.25}{HOOC}-CH_2-CH_2-\underset{\substack{|\\ \overset{\oplus}{N}H_3 \\ pK_a = 9.67}}{CH}-COOH \quad pK_a = 2.19$$

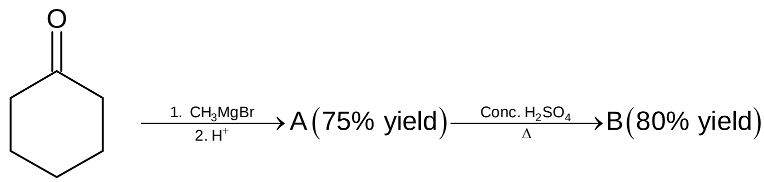
What is the pH corresponding to the isoelectric point of the amino acid?

52. Calculate the volume occupied by 8.8 gm of CO_2 at $31.1^{o}C$ and 1 bar pressure

53. Solid $Ca(HCO_3)_2$ decomposes as

$$Ca(HCO_3)_2(s) \rightleftharpoons CaCO_3(s) + CO_2(g) + H_2O(g)$$

If the total pressure is 0.2 bar, at 420 K, what is standard free energy change of the reaction (in kJ/mole)

54. 

(X)

How many grams of B is obtained (in gram) if initially 9.8 gm X was present?

55. Calculate EMF of the cell in volt.

$\text{Pt} \mid H_2(g) \parallel \text{BOH}(aq) \parallel \text{HA}(aq) \mid H_2(g) \mid \text{Pt}$

$0.1\,\text{bar} \quad 1\,\text{M} \quad 0.1\,\text{M} \quad 1\,\text{bar}$

56. Using van der Waal's equation, find the constant 'a' (in $L^2\,mol^{-2}$) when two moles of a gas is confined in 4 L flask exerts a pressure of 11 atm at a temperature of 300 K?

$\left(b = 0.05\text{ L mol}^{-1}, R = 0.082\text{ L atm / K mol}\right)$

57. A first order reaction is 50% completed in 20 minutes at $27^{o}C$ and in 5 minutes at $47^{o}C$. What is the activation energy of the reaction (in kJ/mole) $[\ell n2 = 0.7]$

58. A solution contains 8.2% Na_3PO_4 and 12% $MgSO_4$ by weight. If percent ionization of Na_3PO_4 and $MgSO_4$ are 50 and 60 respectively, then what is the boiling point $\left(\text{in }^{o}C\right)$?

$\left[K_b(H_2O) = 0.5\text{ kg mol}^{-1}\right]$

59. Molybdenum (At. Wt. = 96 gm/mole) crystallises as bcc. If the density of crystals 10.3 gm / cm^3. What is the radius of Mo atm (in pm)? $\left[N_A = 6\times10^{23}\right]\left[\text{Given}\left(3.1\times10^{-23}\right)^{1/3} = 3.14\times10^{-8}\right]$

60. If the number of equivalent Cr – O bonds in CrO_4^{2-} is x, then $\frac{2x}{3}$?

Mathematics

PART – C

SECTION – A
(One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. The number of integral ordered pairs (x, y) that satisfy the system of equation $|x + y - 4| = 5$ and $|x - 3| + |y - 1| = 5$ is/are
(A) 2
(B) 4
(C) 6
(D) 12

62. The value of $\lim\limits_{n\to\infty}\left(\frac{\lfloor n}{n^n}\right)^{\frac{3n^3+4}{4n^4-1}}$ is
(A) $e^{\frac{3}{4}}$
(B) $e^{-\frac{3}{4}}$
(C) e^{-1}
(D) 0

63. $f(x) = 1 + \int_0^1 \left(xe^y + ye^x\right) f(y)\,dy$, if h(x) = f(x) + 3x is strictly increasing in (–∞, k), then $\left[\frac{4}{3}e^k\right]$ equals to
(where [.] denotes greatest integer function)
(A) 1
(B) 2
(C) 3
(D) $\frac{4}{5}$

64. If a, b $\in$ R distinct numbers satisfying $|a - 1| + |b - 1| = |a| + |b| = |a + 1| + |b + 1|$, then the minimum value of $|a - b|$ is
(A) 1
(B) 2
(C) 3
(D) 4

65. If $a \ne p$, $b \ne q$, $c \ne r$ and $\begin{vmatrix} p & b & c \\ a & q & c \\ a & b & r \end{vmatrix} = 0$, then the value of $\frac{p}{p-a} + \frac{q}{q-b} + \frac{r}{r-c} = ?$
(A) 1
(B) –1
(C) 2
(D) –2

66. A coin is tossed (m + n) times m > n, find the probability of getting atleast m consecutive heads is (m > n)
(A) $\frac{n+2}{m+n}$
(B) $\frac{n+2}{2^m}$
(C) $\frac{n+2}{2^{m+1}}$
(D) $\frac{n+2}{2^n+1}$

67. If α, β, γ are the roots of $2x^3 + x^2 - 7 = 0$, then find the value of $\sum\left(\frac{\alpha}{\beta}+\frac{\beta}{\alpha}\right)$
(A) 3
(B) –3
(C) 2
(D) –2

68. A matrix of order 3×3 are formed by using the elements of the set A = {–3, –2, –1, 0, 1, 2, 3}, then find the probability that the matrix is either symmetric of skew symmetric is
(A) $\frac{1}{7^6}+\frac{1}{7^3}$
(B) $\frac{1}{7^9}+\frac{1}{7^3}-\frac{1}{7^6}$
(C) $\frac{1}{7^3}+\frac{1}{7^6}-\frac{1}{7^9}$
(D) $\frac{1}{7^3}+\frac{1}{7^9}$

69. If $\frac{d}{dx}(f(x))=g(x)$; $\frac{d}{dx}(g(x))=f(x^2)$, then $\frac{d^2}{dx^2}f(x^3)$ can be expressed in the form $kx^a f(x^b)+px\,g(x^x)$ where a, b, c, k, p $\in$ N, then the value of (k + p + a + b + c) equal to
(A) 26
(B) 27
(C) 28
(D) 30

70. If A and B are two square matrix such that $A^2B = BA$ and if $(AB)^{10} = A^kB^{10}$, then k is
(A) 1001
(B) 1023
(C) 1042
(D) 1025

71. Two perpendicular chords of the ellipse $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ (a > b) each passing through exactly one of the foci meet at point P and PQ and PR are tangents to $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, then 4 $\text{cosec}^2(\angle QPR)$ is equal to
(A) 4

(B) 16
(C) 32
(D) 48

72. A circle touches the parabola $y^2 = 4x$ at (1, 2) and also touches it's directrix, then the y-coordinate of the point of contact of the circle and the directrix is
(A) $\sqrt{2}$
(B) 2
(C) $2\sqrt{2}$
(D) 4

73. If x_1, x_2, x_3 are non negative real roots of equation $x^3 - 6x^2 + 3px - 2p = 0$ $(p \in R)$, then value of $\sin^{-1}\left(\frac{1}{x_1}+\frac{1}{x_2}\right)+\cos^{-1}\left(\frac{1}{x_2+x_3}\right)-\tan^{-1}\left(\frac{1}{x_3}+\frac{1}{x_1}\right)$ is equal to
(A) $\frac{\pi}{4}$
(B) $\frac{\pi}{2}$
(C) $\frac{3\pi}{4}$
(D) π

74. $\lim\limits_{n\to\infty}\left[\left((2009)^{2010}\right)^n+\left((2010)^{2009}\right)^n\right]^{\frac{1}{n}}$ is equal to a^b $(a, b \in N)$, then $a - b = ?$
(A) 2009
(B) 1
(C) –1
(D) 0

75. $f : R \to R$ defined as $f(x) = x^3 + 2x^2 + 4x + \sin\frac{\pi x}{2}$ and g be the inverse of f, then $g'(8)$ equals
(A) $\frac{1}{9}$
(B) 9
(C) $\frac{1}{11}$
(D) 11

76. The maximum value of $(\cos\alpha_1)\cdot(\cos\alpha_2)\cdot(\cos\alpha_3)$ ..... $(\cos\alpha_n)$ under the restrictions $0 \le \alpha_1, \alpha_2, \alpha_3 ..... \alpha_n \le \frac{\pi}{2}$ and $\cot\alpha_1 \cdot \cot\alpha_2 \cdot \cot\alpha_3 ..... \cot\alpha_n = 1$ is
(A) $\frac{1}{2^{\frac{n}{2}}}$
(B) $\frac{1}{2^n}$
(C) $\frac{1}{2n}$
(D) 1

77. The X – Y plane is rotated about it's line of intersection with Y – Z plane by 45°, then the equation of new plane is
(A) z – x = 0
(B) x – y = 0
(C) y – z = 0
(D) x + y + z = 0

78. If $\vec{a} \perp r\ \vec{b}$, then vector $\vec{v}$ in terms of $\vec{a}$ and $\vec{b}$ satisfying the condition $\vec{v}\cdot\vec{a}$ and $\vec{v}\cdot\vec{b} = 1$ and $\left[\vec{v}\ \vec{a}\ \vec{b}\right] = 1$ is
(A) $\frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|^2}$
(B) $\frac{\vec{b}}{|\vec{b}|} + \frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|^2}$
(C) $\frac{\vec{b}}{|\vec{b}|^2} + \frac{\vec{a}\times\vec{b}}{|\vec{a}\times\vec{b}|}$
(D) $\vec{b} + \frac{(\vec{a}\times\vec{b})}{|\vec{a}\times\vec{b}|}$

79. Which of the following is not tautology?
(A) $(p \vee q) \rightarrow p$
(B) $(p \vee q) \rightarrow (p \vee \sim q)$
(C) $p \rightarrow (p \vee q)$
(D) $(p \wedge q) \rightarrow (\sim p \vee q)$

80. The mean of two samples of sizes 200 and 300 were found to be 25 and 10 respectively, their standard deviations were 3 and 4 respectively the variance of combined sample size of 500 is
(A) 64
(B) 65.2
(C) 67.2
(D) 64.2

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE.** For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXXX.XX).

81. If $\int \left(x^{2010} + x^{804} + x^{402}\right)\left(2x^{1608} + 5x^{402} + 10\right)^{\frac{1}{402}} = \frac{1}{10a}\left(2x^{2010} + 5x^{804} + 10x^{402}\right)^{\frac{a}{402}}$, then find the value of a

82. The sum of the squares of the lengths of the chords intercepted by the family of the lines x + y = n ($n \in N$) from the circle $x^2 + y^2 = 4$ is

83. Let t and f(t) be the eccentricities of the ellipse $\frac{x^2}{3b^2 - 2a^2} + \frac{y^2}{2b^2 - a^2} = 1$ and $\frac{x^2}{2b^2 - a^2} + \frac{y^2}{b^2} = 1$ where $b^2 > a^2$ ($a \neq 0$) respectively, then $\int_0^{\frac{1}{2}} \text{fofofof}(t)\,dt$ is equal to

84. If $\int_0^1 \left(4x^3 - f(x)\right) f(x)\,dx = \frac{4}{7}$, then find the area of region bounded by y = f(x), x-axis and coordinate x = 1 and x = 2

85. If the sum of first 84 terms of the series $\frac{4+\sqrt{3}}{1+\sqrt{3}} + \frac{8+\sqrt{15}}{\sqrt{3}+\sqrt{5}} + \frac{12+\sqrt{35}}{\sqrt{5}+\sqrt{7}} + ...$ = 549k, then k is equal to

86. The digit at the unit place in the number $(17)^{1995} + (11)^{1995} - (7)^{1995}$

87. If a, b, c $\in$ R and abc = 1 if $A = \begin{bmatrix} 3a & b & c \\ b & 3c & a \\ c & a & 3b \end{bmatrix}$ and $A^T A = 4^{\frac{1}{3}} I$ & |A| > 0, then the value of $a^3 + b^3 + c^3$ is

88. Let circle S_1 and S_2 on argand plane be given by |z + 1| = 3 and |z – 2| = 7 respectively. If a variable circle (S) $|z - z_0| = r$ be inside circle S_2 such that it touches S_1 externally and S_2 internally, then the locus of z_0 describes a conic E. If eccentricity of E can be written in simplest from as $\frac{p}{q}$, then find p + q is (where p, q $\in$ N)

89. A curve y = f(x) is passing through (2, 0) and slope of it's tangent at any point (x, y) is $\frac{(x+1)^2 + y - 3}{x+1}$, then the area bounded by the curve y = f(x) and x-axis is $\frac{m}{n}$ (where HCF of m and n is 1), then value of m + n is

90. Consider two numbers $N_1 = 10^{83}$ and $N_2 = 10^{70}$ the probability that a randomly chosen positive divisor of N_1 is an integer multiple of N_2 is $\frac{1}{9\alpha}$, then α is equal to